Supporting Information

Title: Taxonomical and functional composition of strawberry microbiome is genotype-dependent.

The following Supporting Information is available for this article:

Fig. S1 REP-PCR polyacrylamide gel.

Fig. S2 Rarefaction curves.

Fig. S3 Overall community composition based on relative-abundance data.

Fig. S4 NMDS plots**.**

Fig. S5 *Ps. fluorescens* detection.

Fig. S6 Viable bacterial and fungal cells**.**

Table S1 Strawberry mineral composition.

Table S2 Disease index.

Table S3 List of primers used in this work and their references.

Table S4 Potential strawberry fungal pathogens.

Table S5 Potential beneficial bacterial functions.

Table S6 Relative abundance data statistical analysis.

Table S7 Strawberry chemical composition.

Table S8 Strawberry core.

Table S9 Assigned functions.

Table S10 Beneficial fungal OTUs.

Table S11 Bacterial isolates screened for Plant Growth Promoting traits.

Table S12 Strawberry quality parameters.

Table S13 Nitrogen fixing genera list.

Fig. S1 REP-PCR profiles. Strains in bold represent those selected for identification by 16s sequencing.

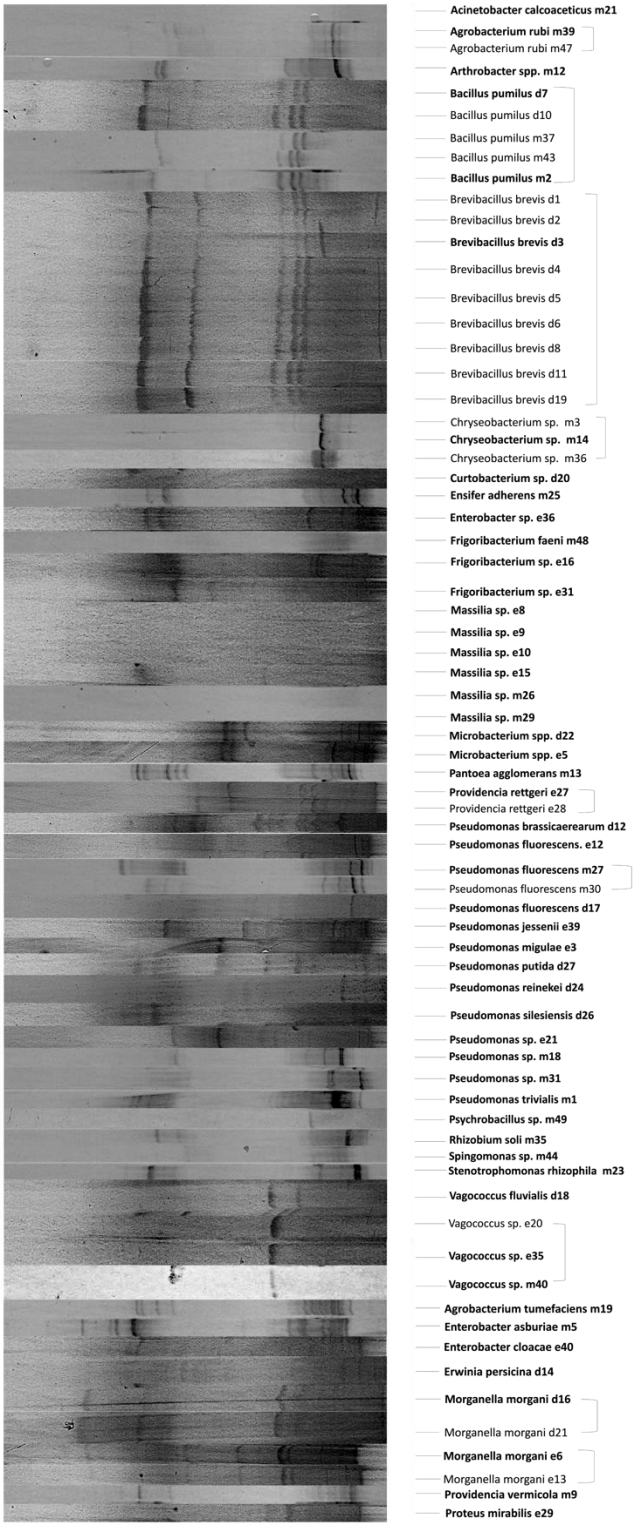

Fig. S2 Rarefaction curves for (A) bacteria and (B) fungi are shown**.**

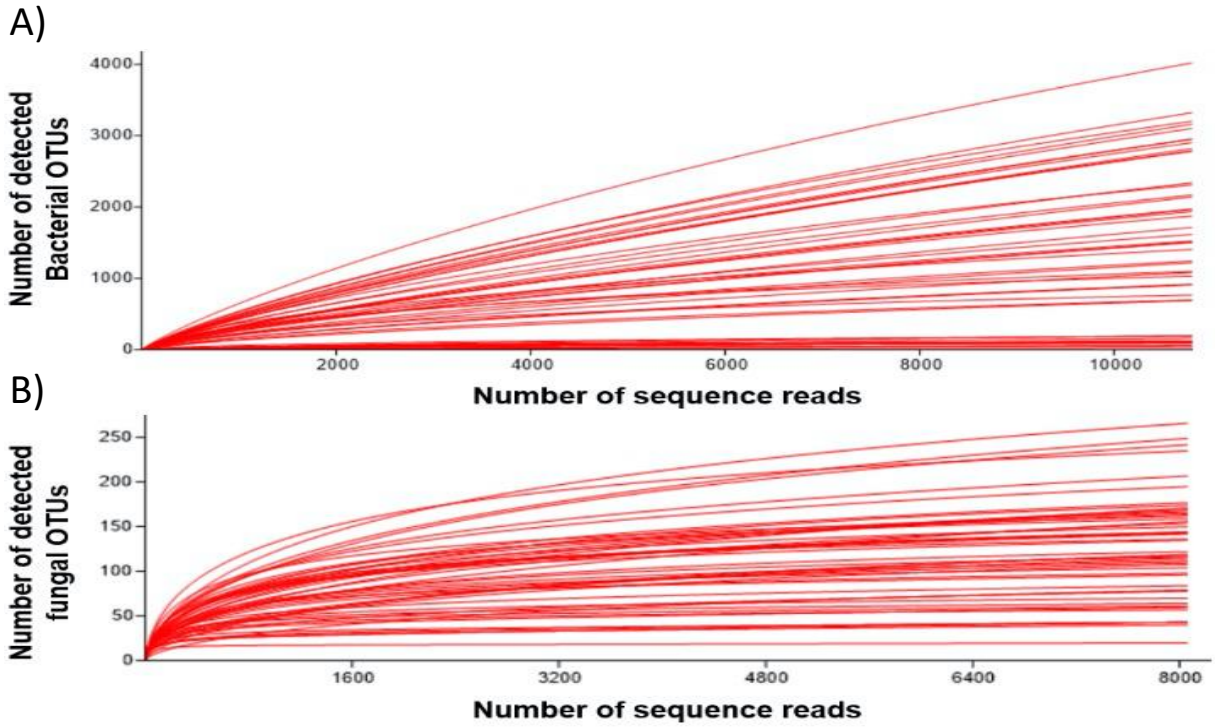

Fig. S3 Overall community composition based on relative-abundance data.

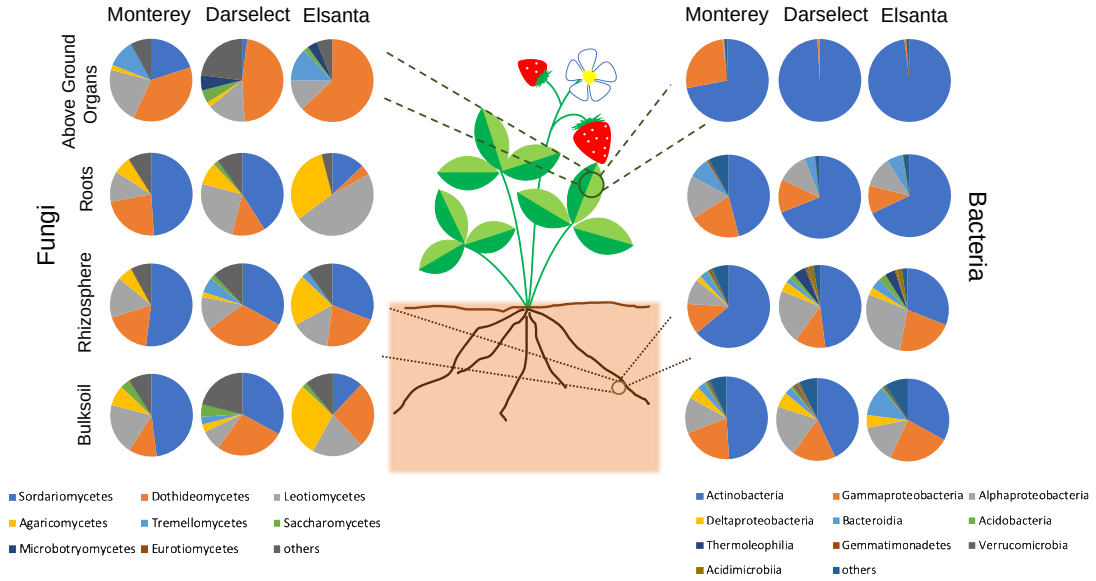

Fig. S4 Nonmetric multidimensional scaling (NMDS) plot showing beneficial community composition similarity of beneficial bacteria (A) and fungi (B) among different soil and plant compartments and genotypes.

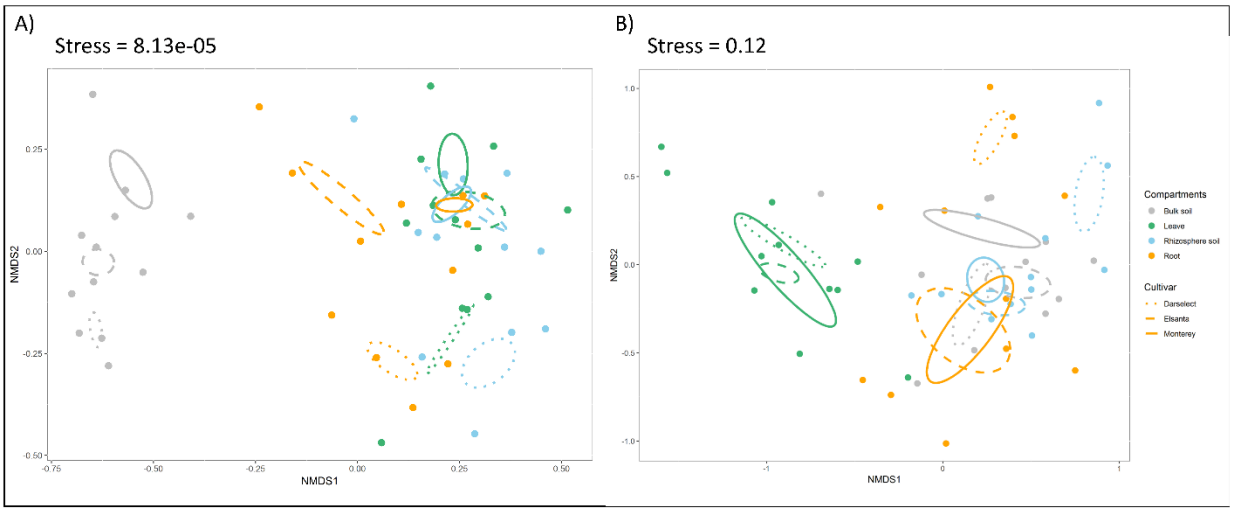

Fig. S5 Gel electrophoresis of PCR amplified products using species-specific primers for *Pseudomonas fluorescens.* Colonization of Monterey genotype has been investigated in bulk soil, rhizosphere, internal tissue of roots and above ground plant compartments; 4 replicates for each organ.

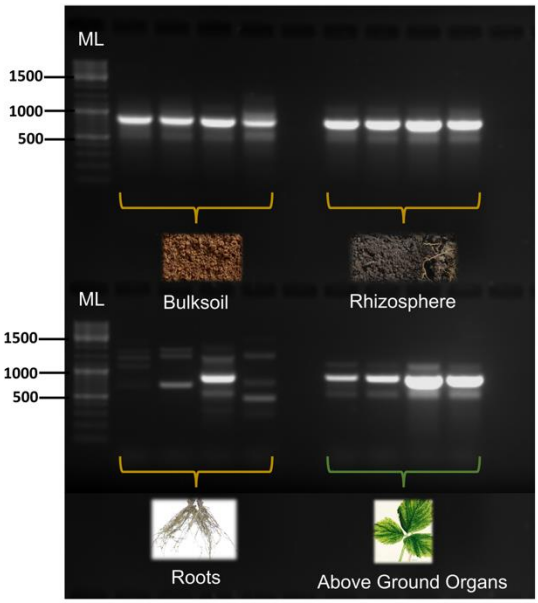

Fig. S6 Boxplots visualizing results for viable bacterial and fungal cells (colony-forming unit, CFU) present in plant and soil compartments of strawberry cvs Darselect, Elsanta, Monterey.

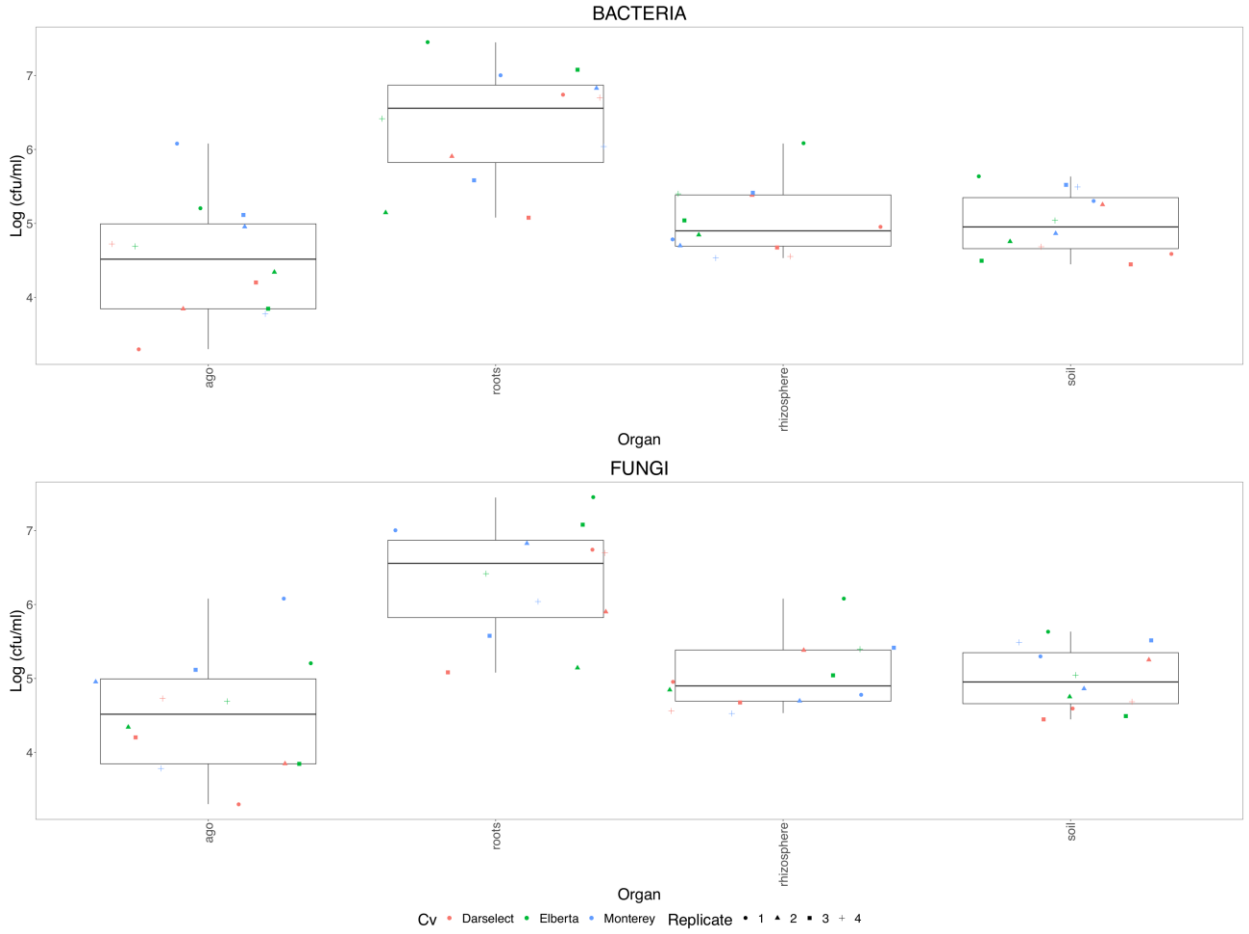

Table S1 Mineral composition of the fertigating solution of strawberry plants in this work.

Macro elements	**mmol**
NO_3	11
NH_4	0.5
H_2PO_4	1
K	6.5
Ca	4
Mg	1.5
SO_4	1.5
Meso elements	**μmol**
Fe	20
$MnSO_4$	15
Micro elements	**μmol**
$ZnSO_4$	5
$Na_2B_4O_7$	4
$CuSO_4$	0.75
Na_2MnO_4	0.5

Table S2 Disease Index (a) of cvs Monterey, Elsanta and Darselect under cultural conditions, based on symptoms incidence and severity determined according to visual scale (b) (0=no symptoms, 5=severe symptoms).

a)

	Monterey	**Elsanta**	**Darselect**
Leaf spot	**0**	**1**	**3**
Powdery mildew	**2**	**5**	**4**

b)

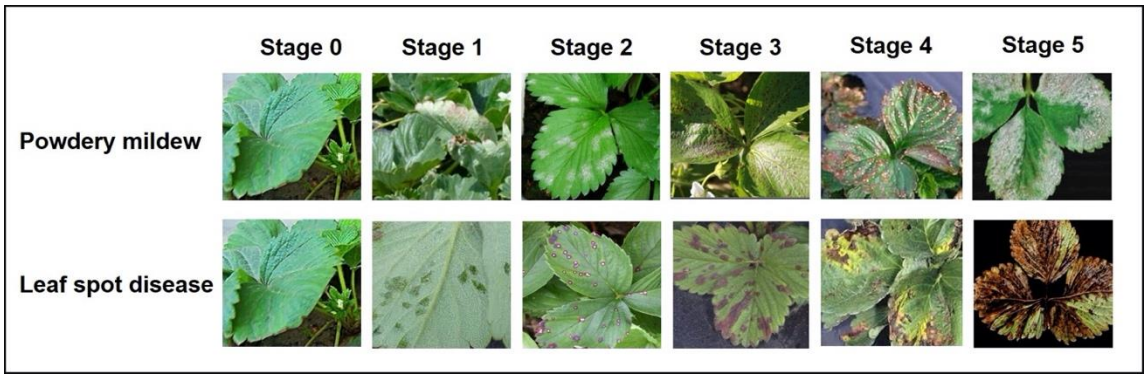

Table S3. List of primers used in this work and their references.

Name	Sequence	Reference
Lac16S for	5’-AATGAGAGTTTGATCCTGGCT-3’	F. Bringel, A. Castioni, D.K. Olukoya, G. E. Felis, S. Torriani, F. Dellaglio. *Lactobacillus plantarum* subsp. argentoratensis subsp. nov., isolated from vegetable matrices. Int. J. System. Evol. Microbiol. 55 (2005) 1629-1634.
Lac16S rev	5’-GAGGTGATCCAGCCGCAGGTT-3’	
i5	5’-TCGTCGGCAGCGTCAGATGTGTATAAGAGACAG-3’	Illumina Primers (Illumina Inc., San Diego, CA, USA)
799F	5’-AACMGGATTAGATACCCKG-3’	M.K. Chelius, E.W. Triplett, The diversity of archaea and bacteria in association with the roots of *Zea mays* L., Microb. Ecol. 41(2011) 252–263. https://doi.org/10.1007/s002480000087
fITS7	5′- GTGARTCATCGAATCTTTG-3′	K. Ihrmark, Katarina, et al., New primers to amplify the fungal ITS2 region–evaluation by 454-sequencing of artificial and natural communities, FEMS microbiology ecology 82.3

Name	Sequence	Reference
		(2012) 666-677. https://doi.org/10.1111/j.1574-6941.2012.01437.x
i7	5’-GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAG-3’	Illumina primers (Illumina Inc., San Diego, CA, USA)
1193r	5’-ACGTCATCCCCACCTTCC-3’	N. Bodenhausen, M.W. Horton, J. Bergelson, Bacterial communities associated with the leaves and the roots of *Arabidopsis thaliana*, PLoS One 8 (2013) e56329. https://doi.org/10.1371/journal.pone.0056329
ITS4	5′-TCCTCCGCTTATTGATATGC-3′	T.J. White TJ, T. Bruns T, S. Lee, J.W. Taylor. Amplification and direct sequencing of fungal ribosomal RNA genes for phylogenetics, San Diego, CA: Academic Press; 1990
i5 index	AATGATACGGCGACCACCGAGATCTACAC[i5 index]TCGTCGGCAGCGTC	Illumina primers (Illumina Inc., San Diego, CA, USA)
i7 index	CAAGCAGAAGACGGCATACGAGAT[i7index]GTCTCGTGGGCTCGG	
PolF	5'- TGC GAY CCS AAR GCB GAC TC -3'	B. Hoppe, T. Kahl, P. Karasch, T. Wubet, J.
PolR	5’- ATS GCC ATC ATY TCR CCG GA -3'	

Name	Sequence	Reference
		Bauhus, F. Buscot, et al., Network analysis reveals ecological links between N-fixing bacteria and wood-decaying fungi, PLoS One 9 (2014) e88141. https://doi.org/10.1371/journal.pone.0088141
16SPS EfluF	5'-TGC ATT CAA AAC TGA CTG-3'	E.S. Boyd, J.W. Peters, New insights into the evolutionary history of biological nitrogen fixation, Front. Microbiol. 4 (2013). https://doi.org/10.3389/fmicb.2013.00201
16SPS ER	5'-AAT CAC ACC GTG GTA ACC G-3'	

Table S4 Potential strawberry fungal pathogens (Unite Species Hypotheses).

OTU	Phylum	UNITE species identification	UNITE species hypotheses (%)	Short ID
Otu0003	Ascomycota	*Botrytis caroliniana*	*Botrytis caroliniana (100)*	SH1189120.08FU
Otu0007	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (100)*	SH1234026.08FU
Otu0016	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (100)*	SH1234026.08FU
Otu0020	Ascomycota	*Ascochyta rabiei*	*Didymella exigua (100)*	SH1174007.08FU
Otu0036	Ascomycota	*Podosphaera sp*	*Podosphaera (100)*	SH1144824.08FU
Otu0040	Ascomycota	*Leptosphaeria sp*	*Paraphoma radicina (100)*	SH1193181.08FU
Otu0053	Ascomycota	*Clonostachys rosea*	*Clonostachys rosea (100)*	SH1155535.08FU
Otu0073	Ascomycota	*Gibellulopsis nigrescens*	*Gibellulopsis piscis (100)*	SH1234027.08FU
Otu0103	Ascomycota	*Stagonosporopsis unclassified*	*Didymella exigua (100)*	SH1174007.08FU
Otu0109	Ascomycota	*Gibberella tricincta*	*Gibberella tricincta (100)*	SH1173588.08FU
Otu0140	Ascomycota	*Mastigosporium album*	*Pezizomycotina (98.64)*	SH1173291.08FU
Otu0145	Ascomycota	*Claviceps sp*	*Hypocreales (100)*	SH1234312.08FU
Otu0160	Ascomycota	*Ramularia sp*	*Ramularia cynarae (100)*	SH1194043.08FU
Otu0169	Ascomycota	*Rhinocladium lesnei*	*Parascedosporium putredinis (100)*	SH1192748.08FU
Otu0177	Ascomycota	*Phaeoacremonium unclassified*	*Sordariomycetes (98.80)*	SH1196099.08FU
Otu0183	Ascomycota	*Ramularia unclassified*	*Ramularia cynarae (100)*	SH1194043.08FU
Otu0230	Ascomycota	*Taphrina wiesneri*	*Taphrina wiesneri (100)*	SH1153415.08FU
Otu0261	Ascomycota	*Taphrina padi*	*Taphrina padi (100)*	SH1153419.08FU
Otu0279	Ascomycota	*Ramularia unclassified*	*Ramularia cynarae (100)*	SH1194043.08FU
Otu0292	Ascomycota	*Seimatosporium sp*	*Xylariales (99.34)*	SH1237195.08FU
Otu0303	Ascomycota	*Trichothecium roseum*	*Trichothecium roseum (100)*	SH1181148.08FU
Otu0325	Ascomycota	*Gibberella pulicaris*	*Fusarium venenatum (100)*	SH1212037.08FU
Otu0355	Ascomycota	*Pestalotiopsis sp*	*Neopestalotiopsis steyaertii (100)*	SH1177548.08FU
Otu0364	Ascomycota	*Cylindrocladiella pseudoparva*	*Cylindrocladiella variabilis (100)*	SH1212036.08FU
Otu0380	Chytridiomycota	*Powellomyces hirtus*	*Powellomyces hirtus (100)*	SH1187133.08FU
Otu0397	Ascomycota	*Neodeightonia phoenicum*	*Neodeightonia palmicola (100)*	SH1143975.08FU
Otu0408	Ascomycota	*Scopulariopsis brumptii*	*Microascus paisii (100)*	SH1146112.08FU
Otu0421	Ascomycota	*Monographella nivalis*	*Monographella nivalis (100)*	SH1179475.08FU
Otu0425	Ascomycota	*Gibellulopsis chrysanthemi*	*Gibellulopsis chrysanthemi (94.67)*	SH1234048.08FU
Otu0427	Ascomycota	*Rhizosphaera kalkhoffii*	*Scleroconidioma sphagnicola (100)*	SH1149663.08FU
Otu0451	Ascomycota	*Taphrina inositophila*	*Taphrina inositophila (98.50)*	SH1153418.08FU
Otu0459	Ascomycota	*Plectosphaerella unclassified*	*Acremonium nepalense (100)*	SH1234026.08FU
Otu0467	Ascomycota	*Gibberella tricincta*	*Gibberella tricincta (100)*	SH1173588.08FU
Otu0479	Ascomycota	*Taphrina pruni*	*Taphrina pruni (99.53)*	SH1153426.08FU
Otu0499	Basidiomycota	*Gjaerumia minor*	*Gjaerumia minor (99.55)*	SH1186349.08FU
Otu0507	Ascomycota	*Blumeria graminis*	*Blumeria graminis (100)*	SH1159375.08FU
Otu0516	Ascomycota	*Bipolaris cynodontis*	*Bipolaris peregianensis (100)*	SH1157992.08FU

OTU	Phylum	UNITE species identification	UNITE species hypotheses (%)	Short ID
Otu0588	Ascomycota	*Devriesia sp*	*Devriesia (100)*	SH1231801.08FU
Otu0601	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (95.86)*	SH1234026.08FU
Otu0612	Ascomycota	*Dendryphion europaeum*	*Dendryphion europaeum (100)*	SH1191756.08FU
Otu0615	Ascomycota	*Stemphylium vesicarium*	*Stemphylium vesicarium (100)*	SH1157993.08FU
Otu0645	Ascomycota	*Mycosphaerella ellipsoidea*	*Dothistroma pini (100)*	SH1194045.08FU
Otu0657	Ascomycota	*Botrytis caroliniana*	*Botrytis caroliniana (97.95)*	SH1189120.08FU
Otu0661	Ascomycota	*Eutypa maura*	*Diatrypaceae (100)*	SH1190467.08FU
Otu0664	Ascomycota	*Trichothecium crotocinigenum*	*Trichothecium crotocinigenum (98.31)*	SH1181149.08FU
Otu0688	Basidiomycota	*Typhula sp*	*Typhula (98.67)*	SH1238811.08FU
Otu0700	Ascomycota	*Golovinomyces sordidus*	*Golovinomyces leucheriae (100)*	SH1210305.08FU
Otu0734	Ascomycota	*Ceratocystidaceae sp*	*Thielaviopsis cerberus (100)*	SH1227619.08FU
Otu0736	Ascomycota	*Pilidium concavum*	*Pilidium concavum (100)*	SH1213824.08FU
Otu0757	Ascomycota	*Taphrina carpini*	*Taphrina carpini (100)*	SH1153417.08FU
Otu0760	Ascomycota	*Pyrenophora teres*	*Pleosporaceae (100)*	SH1192922.08FU
Otu0826	Ascomycota	*Taphrina unclassified*	*Taphrina inositophila (95.50)*	SH1153418.08FU
Otu0842	Basidiomycota	*Melampsora epitea*	*Melampsora epitea (100)*	SH1190669.08FU
Otu0877	Ascomycota	*Stagonosporopsis sp*	*Didymella exigua (98.72)*	SH1174007.08FU
Otu0907	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (100)*	SH1234026.08FU
Otu0942	Ascomycota	*Devriesia thermodurans*		
Otu0983	Basidiomycota	*Thecaphora melandrii*	*Thecaphora saponariae (100)*	SH1142764.08FU
Otu0989	Ascomycota	*Venturia hystrioides*	*Fusicladium fagi (98.62)*	SH1222281.08FU
Otu1117	Ascomycota	*Ascochyta medicaginicola*	*Pleosporales (100)*	SH1213830.08FU
Otu1125	Ascomycota	*Edenia gomezpompae*	*Pezizomycotina (100)*	SH1146626.08FU
Otu1141	Basidiomycota	*Ustilago hordei*	*Ustilago hordei (97.26)*	SH1145429.08FU
Otu1204	Ascomycota	*Gibellulopsis nigrescens*	*Gibellulopsis piscis (98.24)*	SH1234027.08FU
Otu1206	Ascomycota	*Neofusicoccum parvum*	*Neofusicoccum mangiferae (99.35)*	SH1143963.08FU
Otu1257	Ascomycota	*Phaeoacremonium parasiticum*	*Phaeoacremonium rubrigenum (100)*	SH1196092.08FU
Otu1281	Ascomycota	*Lectera colletotrichoides*	*Lectera longa (100)*	SH1234028.08FU
Otu1284	Basidiomycota	*Moesziomyces aphidis*	*Moesziomyces parantarcticus (100)*	SH1145427.08FU
Otu1321	Ascomycota	*Rhexocercosporidium panacis*	*Rhynchosporium commune (98.64)*	SH1173280.08FU
Otu1330	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (100)*	SH1234026.08FU
Otu1338	Ascomycota	*Glomerella acutata*	*Colletotrichum acerbum (98.69)*	SH1171875.08FU
Otu1400	Ascomycota	*Hortaea werneckii*	*Hortaea werneckii (100)*	SH1154533.08FU
Otu1431	Ascomycota	*Sclerotinia sclerotiorum*	*Botrytis caroliniana (100)*	SH1189120.08FU
Otu1437	Basidiomycota	*Meira geulakonigii*	*Meira geulakonigii (99.56)*	SH1237725.08FU
Otu1524	Ascomycota	*Clonostachys rosea*	*Clonostachys rosea (90.91)*	SH1155535.08FU
Otu1531	Ascomycota	*Bipolaris sorokiniana*	*Bipolaris peregianensis (100)*	SH1157992.08FU
Otu1577	Ascomycota	*Clonostachys rosea*	*Clonostachys rosea (100)*	SH1155535.08FU

OTU	Phylum	UNITE species identification	UNITE species hypotheses (%)	Short ID
Otu1694	Ascomycota	*Gibberella tricincta*	*Gibberella tricincta (100)*	SH1173588.08FU
Otu1709	Ascomycota	*Botrytis caroliniana*	*Botrytis caroliniana (100)*	SH1189120.08FU
Otu1713	Ascomycota	*Curvularia americana*	*Curvularia lunata (100)*	SH1157991.08FU
Otu1731	Ascomycota	*Rhexocercosporidium panacis*	*Rhynchosporium commune (97.26)*	SH1173280.08FU
Otu1756	Ascomycota	*Clonostachys rosea*	*Clonostachys rosea (97.58)*	SH1155535.08FU
Otu1761	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (100)*	SH1234026.08FU
Otu1781	Ascomycota	*Erysiphe arcuata*	*Erysiphe arcuata (99.45)*	SH1184259.08FU
Otu1821	Ascomycota	*Leptosphaeria sp*	*Paraphoma radicina (100)*	SH1193181.08FU
Otu1841	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (99.31)*	SH1234026.08FU
Otu1866	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (97.63)*	SH1234026.08FU
Otu1875	Ascomycota	*Clonostachys rosea*	*Clonostachys rosea (100)*	SH1155535.08FU
Otu1881	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (98.19)*	SH1234026.08FU
Otu1896	Ascomycota	*Erysiphaceae unclassified*	*Erysiphe (100)*	SH1184255.08FU
Otu1906	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (99.31)*	SH1234026.08FU
Otu1909	Ascomycota	*Leptosphaeria sp*	*Paraphoma radicina (98)*	SH1193181.08FU
Otu1927	Ascomycota	*Pyrenophora avenae*	*Pleosporaceae (100)*	SH1192921.08FU
Otu1930	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium furcatum (99.42)*	SH1234029.08FU
Otu2003	Ascomycota	*Botrytis caroliniana*	*Botrytis caroliniana (96.25)*	SH1189120.08FU
Otu2023	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (100)*	SH1234026.08FU
Otu2033	Ascomycota	*Botrytis caroliniana*	*Botrytis caroliniana (100)*	SH1189120.08FU
Otu2059	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (97.63)*	SH1234026.08FU
Otu2086	Ascomycota	*Plectosphaerella unclassified*	*Acremonium nepalense (99.41)*	SH1234026.08FU
Otu2119	Ascomycota	*Leptosphaeria unclassified*	*Paraphoma radicina (100)*	SH1193181.08FU
Otu2135	Ascomycota	*Glomerella acutata*	*Colletotrichum acerbum (98.66)*	SH1171875.08FU
Otu2146	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (99.32)*	SH1234026.08FU
Otu2150	Ascomycota	*Glomerella acutata*	*Colletotrichum acerbum (98..08)*	SH1171875.08FU
Otu2197	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (98.22)*	SH1234026.08FU
Otu2226	Ascomycota	*Ramularia eucalypti*	*Ramularia cynarae (97.87)*	SH1194043.08FU
Otu2233	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense*	SH1234026.08FU
Otu2240	Ascomycota	*Gibberella unclassified*	*Gibberella fujikuroi (98.65)*	SH1212030.08FU
Otu2314	Ascomycota	*Gibellulopsis nigrescens*	*Gibellulopsis piscis (95.88)*	SH1234027.08FU
Otu2360	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (99.35)*	SH1234026.08FU
Otu2365	Ascomycota	*Taphrina mirabilis*	*Taphrina pruni-subcordatae (99.53)*	SH1153416.08FU
Otu2369	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (97.63)*	SH1234026.08FU
Otu2382	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (98.22)*	SH1234026.08FU
Otu2384	Ascomycota	*Gibellulopsis nigrescens*	*Gibellulopsis piscis (100)*	SH1234027.08FU
Otu2407	Ascomycota	*Ramichloridium indicum*	*Ramichloridium indicum (100)*	SH1177266.08FU
Otu2423	Ascomycota	*Lepteutypa fuckelii*	*Lepteutypa fuckelii (92.40)*	SH1184890.08FU
Otu2454	Ascomycota	*Plectosphaerella cucumerina*	*Acremonium nepalense (97.63)*	SH1234026.08FU

OTU	Phylum	UNITE species identification	UNITE species hypotheses (%)	Short ID
Otu2490	Basidiomycota	*Urocystis agropyri*	*Urocystis agropyri (98.85)*	SH1177340.08FU
Otu2527	Ascomycota	*Leptosphaeria sp*		

Table S5 Identification of potential bacterial beneficial functions based on literature research.

Function	Taxa
Nitrogen fixation	*Allorhizobium, Bradyrhizobium, Ensifer, Mesorhizobium, Rhizobium, Neorhizobium, Pararhizobium, Aminobacter, Devosia, Methylobacterium, Microvirga, Phyllobacterium, Shinella, Burkholderia, Paraburkholderia, Cupriavidus* (Khan et al., 2010)
Plant growth promotion and Biological control activity	*Bacillus megaterium* and *B. pumilus* (Choudhary and Johri, 2009)
	Cellulosimicrobium sp. (Nabti et al., 2014)
	Pseudomonas fluorescens (Purahong et al., 2018)
	Streptomyces hygroscopicus (Prapagdee et al., 2008)

Table S6 Effect of genotype, soil and plant compartment on community composition of strawberry microbiome based on relative abundance data and Bray-Curtis distance measure. Nd = not determined; Significant *P* values are highlighted in bold.

Microorganisms/Factors	Community composition (Two-way ANOSIM)		Community composition (Two-way PERMANOVA)	
	R	***P***	**Pseudo*F***	***P***
Total bacteria				
Genotype	0.3	**0.001**	3.19	**0.003**
Compartment	0.54	**0.001**	10.43	**0.001**
Genotype x compartment	nd	nd	1.97	**0.005**
Potential beneficial bacteria				
Genotype	0.30	**0.001**	2.05	**0.001**
Compartment	0.48	**0.001**	5.69	**0.001**
Genotype x compartment	nd	nd	1.47	**0.002**
Fungi				
Genotype	0.45	**0.001**	4.14	**0.001**
Compartment	0.65	**0.001**	7.77	**0.001**
Genotype x compartment	nd	nd	1.92	**0.002**
Potential beneficial fungi				
Genotype	0.2	**0.001**	2.4926	**0.001**
Compartment	0.53	**0.001**	9.8272	**0.001**
Genotype x compartment	nd	nd	1.8785	**0.005**
Plant pathogenic fungi				
Genotype	0.36	**0.001**	3.6854	**0.001**
Compartment	0.45	**0.001**	5.2552	**0.001**
Genotype x compartment	nd	nd	1.9834	**0.001**

Table S7 Chemical composition of roots and above-ground strawberry organs.

	Above-ground Organs			Roots		
	Darselect	**Elsanta**	**Monterey**	**Darselect**	**Elsanta**	**Monterey**
Macro-elements, % dry weight						
C	43.50 ± 0.82	44.40 ± 0.98	43.11 ± 0.83	46.20 ± 3.13	48.25 ± 0.54	45.77 ± 0.58
N	3.25 ± 0.43	3.02 ± 0.34	3.36 ± 0.49	1.36 ± 0.10	1.09 ± 0.15	1.60 ± 0.11
H	5.75 ± 0.12	5.73 ± 0.14	5.52 ± 0.13	5.10 ± 0.23	5.23 ± 0.04	5.24 ± 0.21
Macro-elements, µg/g						
P	4425.60 ± 1021.05	3099.75 ± 905.37	5598.78 ± 528.59	1285.95 ± 198.23	752.55 ± 124.10	1544.53 ± 431.20
K	31527 ± 5251.49	17834.63 ± 4314.25	30119.10 ± 1284.10	1158.75 ± 249.05	822.30 ± 424.05	1657.15 ± 1271.44
Ca	14685.33 ± 1708.42	11495.10 ± 1281.37	10188.78 ± 978.89	11636.68 ± 2139.82	12512.58 ± 1943.92	17586.35 ± 1616.47
Mg	4117.93 ± 615.78	4675 ± 731.02	5103.70 ± 669.97	2600.35 ± 346.89	2174.43 ± 246.41	3638.43 ± 579.04
Micro-elements, µg/g						
B	59.90 ± 6.93	65.18 ± 19.91	67.65 ± 9.26	24.53 ± 3.66	19.33 ± 0.56	33.18 ± 2.55
Mn	32.88 ± 5.39	86.88 ± 55.97	41.20 ± 15.08	45.43 ± 31.63	35.83 ± 24.36	74.75 ± 23.16
Fe	71.08 ± 10.45	89.85 ± 18.70	106.68 ± 17.44	517.95 ± 281.97	218.08 ± 73.42	533.83 ± 387.58
Co	0.00 ± 0.00	0.03 ± 0.05	0.00 ± 0.00	0.25 ± 0.13	0.20 ± 0.00	0.48 ± 0.42
Ni	0.30 ± 0.14	0.45 ± 0.24	0.30 ± 0.00	0.90 ± 0.37	0.65 ± 0.13	1.83 ± 0.77
Zn	23.43 ± 3.29	21.70 ± 9.35	29.60 ± 7.66	81.73 ± 28.97	39.68 ± 18.35	26.58 ± 3.01
Sr	93.28 ± 14.46	77.08 ± 10.84	63.88 ± 7.85	82.70 ± 15.55	81.25 ± 10.09	132.40 ± 5.35
Mo	0.95 ± 0.37	1.08 ± 0.36	2.25 ± 0.44	14.33 ± 5.94	19.40 ± 15.72	24.90 ± 7.58
Ba	24.68 ± 4.10	22.68 ± 4.77	16.78 ± 4.74	15.93 ± 4.35	9.80 ± 1.39	31.53 ± 9.11
Pb	0.23 ± 0.05	0.13 ± 0.05	0.18 ± 0.10	3.98 ± 1.89	2.53 ± 0.50	1.35 ± 0.57

Table S8 Bacterial (mainly Micrococcales) and fungal (mainly Ascomycota) OTUs detected in all 3 strawberry genotypes and in all plant-soil compartments.

OTU	Phylum	Order	Family	Genus	Species
Otu000001	Actinobacteria	Micrococcales	Micrococcaceae	*Pseudarthrobacter*	-
Otu000003	Gammaproteobacteria	Pseudomonadales	Pseudomonadaceae	*Pseudomonas*	-
Otu000005	Actinobacteria	Micrococcales	Micrococcaceae	*Paenarthrobacter*	-
Otu000007	Gammaproteobacteria	Betaproteobacteriales	Burkholderiaceae	*Limnohabitans*	-
Otu000010	Actinobacteria	Micrococcales	Micrococcaceae	-	-
Otu000009	Actinobacteria	Micrococcales	Micrococcaceae	*Paenarthrobacter*	-
Otu000014	Actinobacteria	Micrococcales	Micrococcaceae	*Paenarthrobacter*	-
Otu000016	Actinobacteria	Micrococcales	Micrococcaceae	*Paenarthrobacter*	-
Otu000024	Actinobacteria	Micrococcales	Micrococcaceae	*Paenarthrobacter*	-
Otu000017	Actinobacteria	Micrococcales	Micrococcaceae	-	-
Otu000018	Actinobacteria	Micrococcales	Micrococcaceae	-	-
Otu000046	Actinobacteria	Micrococcales	Micrococcaceae	*Paenarthrobacter*	-
Otu000043	Actinobacteria	Micrococcales	Micrococcaceae	*Paenarthrobacter*	-
Otu000060	Actinobacteria	Micrococcales	Micrococcaceae	-	-
Otu000032	Actinobacteria	Micrococcales	Micrococcaceae	*Paenarthrobacter*	-
Otu000066	Actinobacteria	Micrococcales	Microbacteriaceae	-	-
Otu000084	Actinobacteria	Micrococcales	Micrococcaceae	-	-
Otu000090	Actinobacteria	Micrococcales	Micrococcaceae	-	-
Otu000120	Actinobacteria	Micrococcales	Micrococcaceae	-	-
Otu000100	Actinobacteria	Micrococcales	-	-	-
Otu000171	Actinobacteria	Micrococcales	Micrococcaceae	-	-
Otu000193	Actinobacteria	Micrococcales	Microbacteriaceae	-	-
Otu000279	Actinobacteria	Micrococcales	Micrococcaceae	-	-
Otu000241	Actinobacteria	Micrococcales	Micrococcaceae	*Paenarthrobacter*	-
Otu0001	Ascomycota	Capnodiales	-	-	*Capnodiales unclassified*
Otu0002	Ascomycota	Helotiales	-	*Cadophora*	*Cadophora luteo-olivacea*
Otu0003	Ascomycota	Helotiales	Sclerotiniaceae	*Botrytis*	*Botrytis caroliniana*
Otu0007	Ascomycota	Glomerellales	Plectosphaerellaceae	*Plectosphaerella*	*Plectosphaerella cucumerina*
Otu0008	Fungi_unclassified	-	-	-	-
Otu0011	Ascomycota	Saccharomycetales	Saccharomycetaceae	*Saccharomyces*	*Saccharomyces cerevisiae*
Otu0017	Ascomycota	Dothideales	Aureobasidiaceae	*Aureobasidium*	*Aureobasidium pullulans*
Otu0018	Ascomycota	Capnodiales	Cladosporiaceae	*Cladosporium*	*Cladosporium sphaerospermum*
Otu0020	Ascomycota	Pleosporales	Didymellaceae	*Ascochyta*	*Ascochyta rabiei*

OTU	**Phylum**	**Order**	**Family**	**Genus**	**Species**
Otu0022	Basidiomycota	Tremellales	Bulleribasidiaceae	*Vishniacozyma*	*Vishniacozyma victoriae*
Otu0025	Ascomycota	Pleosporales	Pleosporaceae	*Alternaria*	*Alternaria alternata*
Otu0043	Basidiomycota	Holtermanniales		*Holtermanniella*	*Holtermanniella takashimae*
Otu0045	Ascomycota	Pleosporales	Pleosporaceae	*Alternaria*	-
Otu0099	Ascomycota	Hypocreales	Cordycipitaceae	*Lecanicillium*	*Lecanicillium muscarium*
Otu0135	Basidiomycota	Tremellales	Bulleribasidiaceae	*Vishniacozyma*	*Vishniacozyma heimaeyensis*

Table S9 Number of OTUs belonging to particular bacterial or fungal functions assigned based on FAPROTAX and FUNguild, respectively.

BACTERIA - FUNCTIONS	OTUs
Aromatic compound degradation	99
Cellulolysis	31
Chemoheterotrophy	2389
Chitynolysis	3
Hydrocarbon degradation	41
Intracelluar parasites	235
Iron respiration	2
Manganese oxidation	12
Methanol oxidation	444
Multifunctions	231
Nitrate reduction	40
Nitrite oxidation	24
Nitrogen fixation	19
Photoautotrophy	27
Photoheterotrophy	8
Predatory/exoparasitic	233
Sulfate respiration	27
Sulfur compound oxidation	82
Ureolysis	126
Xylanolysis	2
Total assigned functions	*20*
FUNGI - FUNCTION	
Animal endosymbiont-saprotroph	1
Animal parasite-fungal parasite	1
Animal pathogen	34
Arbuscular mycorrhizal[a]	6
Ectomycorrhizal[a]	5
Endophyte[a]	57
Endophyte-plant pathogen	7
Endophyte-saprotroph	9
Ericoid mycorrhizal (dark septate endophyte)[a]	22
Fungal parasite[a]	13
Lichen parasite	1
Lichenized	1
Lichenized-saprotroph	1
Multifunctions	55
Plant pathogen[b]	136
Saprotroph	357
Total assigned functions	*16*

Fungal OTU classification: a= beneficial, b=pathogenic.

Table S10 List of fungal OTUs identified at the species level, for which biological control or plant growth promoting activities have been described in literature.

Activities	Fungal species
BCA against fungal pathogens	*Acremonium alternatum* (Raps and Vidal, 1998)
	Aureobasidium pullulans (Mari et al., 2012)
	Candida sake (Marìn et al., 2016)
	Chaetomium globosum (Shanthiyaa et al., 2013)
	Chaetomium cupreum (Soytong et al., 2001)
	Clonostachys rosea (Ravnskov et al., 2006)
	Coprinopsis urticicola (Gholami et al., 2019)
	Debaryomyces hansenii (Gil-Serna et al., 2011)
	Mortierella hyaline (Gholami et al., 2019)
	Papiliotrema flavescens (Schisler et al., 2019)
	Rhodotorula babjevae (Sandberg, 2019)
	Ceriporia lacerata (Yin et al., 2018)
	Chaetomium globosum (Khan et al., 2012)
	Penicillium adametzii (Szwajkowska-Michalek et al., 2012)
	Saccharomyces cerevisiae (Lopes et al., 2015)
Induction of systemic resistance	*Trichoderma harzianum* (Adnan et al., 2019)
Entomopathogenic activity	*Beauveria bassiana* (Zimmermann, 2007)
	Lecanicillium muscarium (Ownley et al., 2010)
Phytase production	*Chaetomium globosum* (Shanthiyaa et al., 2013; Soytong et al., 2001)
Improvement of Soil Quality by Production of Chitins and Phytohormones	*Chaetomium globosum* (Shanthiyaa et al., 2013; Soytong et al., 2001)
	Aspergillus fumigatus (Khan et al., 2011)
	Cladosporium sphaerospermum Hamayun et al., 2010)
	Mortierella elongate (Li et al., 2018)
	Fusarium proliferatum (Bilal et al., 2018)
Plant Growth Promotion by enhancement of mineral nutrients use efficiency	*Periconia macrospinosa* (Yakti et al., 2018)
Improvement of plant Growth under saline stress by enhancement of antioxidative defense system	*Trichoderma longibrachiatum* (Zhang et al., 2016)

Table S11 Bacterial isolates screened for Plant Growth Promoting traits. See attached excel file.

Table S12 Quality parameters evaluated on harvested strawberry fruits.

Cultivar	Diameter (mm)	Texture (N)	pH	SSC (°Brix)	Titratable acidity (citric acid eqs, mg/ml)
Darselect	34.57±1.99	2.29±0.20	3.75±0.03	6.37±1.23	3.71±1.88
Elsanta	33.03±3.06	2.20±0.46	3.68±0.11	11.55±3.23	3.94±1.22
Monterey	31.26±4.82	3.05±1.28	3.61±0.05	5.67±2.39	2.70±1.28

Table S13 Nitrogen fixing bacterial genera detected in cultivated strawberry, wild strawberries and legume plants.

Bacterium genera	Cultivated Strawberry	Wild Strawberry	Legumes
Allorhizobium	✓ (71 OTUs)	×	✓
Neorhizobium		×	✓
Pararhizobium		×	✓
Rhizobium		✓ (3 OTUs)	✓
Aminobacter	✓ (1 OTU)	✓ (1 OTU)	✓
Azorhizobium	×	×	✓
Bradyrhizobium	✓ (22 OTUs)	✓ (6 OTUs)	✓
Burkholderia	✓ (48 OTUs)	✓ (5 OTU)	✓
Paraburkholderia		×	✓
Cupriavidus	✓ (1 OTU)	×	✓
Devosia	✓ (83 OTUs)	✓ (3 OTU)	✓
Ensifer	✓ (4 OTUs)	×	✓
Mesorhizobium	✓ (30 OTUs)	✓ (1OTU)	✓
Methylobacterium	✓(6 OTUs)	✓ (2 OTU)	✓
Microvirga	✓(3 OTUs)	×	✓
Ochrobactrum	×	×	✓
Phyllobacterium	✓(1 OTU)	×	✓
Shinella	✓(29 OTUs)	×	✓

References cited in Supplementary Material:

- M. Adnan, W. Islam, A. Shabbir, K.A. Khan, H.A. Ghramh, Z. Huang, et al., Plant defense against fungal pathogens by antagonistic fungi with *Trichoderma* in focus, Microb. Pathog. 129 (2019) 7-18. https://doi.org/10.1016/j.micpath.2019.01.042
- L. Bilal, S. Asa, M. Hamayun, H. Gul, A. Iqbal, I. Ullah, et al., Plant growth promoting endophytic fungi *Aspergillus fumigatus* TS1 and *Fusarium proliferatum* BRL1 produce gibberellins and regulates plant endogenous hormones, Symbiosis 76 (2018) 117-127. https://doi.org/10.1007/s13199-018-0545-4
- D.K. Choudhary, B.N. Johri, Interactions of *Bacillus* spp. and plants–with special reference to induced systemic resistance (ISR), Microbiol. Res. 164 (2009) 493-513. https://doi.org/10.1016/j.micres.2008.08.007
- M. Gholami, J. Amini, J. Abdollahzadeh, M. Ashengroph, Basidiomycetes fungi as biocontrol agents against take-all disease of wheat, Biolog. Control 130 (2019) 34-43. https://doi.org/10.1016/j.biocontrol.2018.12.012
- J. Gil-Serna, B. Patiño, L. Cortés, M.T. González-Jaén, C. Vázquez, Mechanisms involved in reduction of ochratoxin A produced by *Aspergillus westerdijkiae* using *Debaryomyces hansenii* CYC 1244, Int. J. Food Microbiol. 151 (2011) 113-118. https://doi.org/10.1016/j.ijfoodmicro.2011.08.012
- M. Hamayun, S.A. Khan, A.L. Khan, G. Rehman, Y.H. Kim, I. Iqbal, et al., Gibberellin production and plant growth promotion from pure cultures of *Cladosporium* sp. MH-6 isolated from cucumber (*Cucumis sativus* L.), Mycologia 102 (2010) 989-995. https://doi.org/10.3852/09-261
- A.L. Khan, M. Hamayun, Y.H. Kim, S.M. Kang, J.H. Lee, I.J. Lee, Gibberellins producing endophytic *Aspergillus fumigatus* sp. LH02 influenced endogenous phytohormonal levels, isoflavonoids production and plant growth in salinity stress, Process Biochem. 46 (2011) 440-447. https://doi.org/10.1016/j.procbio.2010.09.013
- A.L. Khan, Z.K. Shinwari, Y.H. Kim, M. Waqas, M. Hamayun, M. Kamran, et al., Role of endophyte *Chaetomium globosum* LK4 in growth of *Capsicum annuum* by production of gibberellins and indole acetic acid, Pak. J. Bot. 44 (2012) 1601-1607.

- M.S. Khan, A. Zaidi, J. Musarrat, Microbes for legume improvement, Vienna: Springer (2010)
- U. Kõljalg, R.H. Nilsson, K. Abarenkov, L. Tedersoo, A.F. Taylor, M. Bahram, Towards a unified paradigm for sequence-based identification of fungi, Mol. Ecol. 22 (2013) 5271-5277. https://doi.org/10.1111/mec.12481
- F. Li, L. Chen, M. Redmile-Gordon, J. Zhang, C. Zhang, Q. Ning, et al., *Mortierella elongata*'s roles in organic agriculture and crop growth promotion in a mineral soil, Land Degrad. Dev. 29 (2018) 1642-1651. https://doi.org/10.1002/ldr.2965
- M.R. Lopes, M.N. Klein, L.P. Ferraz, A.C. da Silva, K.C. Kupper, Saccharomyces cerevisiae: a novel and efficient biological control agent for *Colletotrichum acutatum* during pre-harvest, Microbiol. Res. 175 (2015) 93-99. https://doi.org/10.1016/j.micres.2015.04.003
- M. Mari, C. Martini, A. Spadoni, W. Rouissi, P. Bertolini, Biocontrol of apple postharvest decay by *Aureobasidium pullulans*, Postharvest Biol. Tec. 73 (2012) 56-62. https://doi.org/10.1016/j.postharvbio.2012.05.014
- A. Marín, M. Cháfer, L. Atarés, A. Chiralt, R. Torres, J. Usall, et al., Effect of different coating-forming agents on the efficacy of the biocontrol agent *Candida sake* CPA-1 for control of *Botrytis cinerea* on grapes, Biol. Control 96 (2016) 108-119. https://doi.org/10.1016/j.biocontrol.2016.02.012
- C. Merchante, J.C. Vallarino, S. Osorio, I. Aragüez, N. Villarreal, M.T. Ariza, et al., Ethylene is involved in strawberry fruit ripening in an organ-specific manner, J. Exp. Botany 64 (2013) 4421-4439. https://doi.org/10.1093/jxb/ert257
- E. Nabti, L. Bensidhoum, N. Tabli, D. Dahel, A. Weiss, M. Rothballer, et al., Growth stimulation of barley and biocontrol effect on plant pathogenic fungi by a *Cellulosimicrobium* sp. strain isolated from salt-affected rhizosphere soil in northwestern Algeria, Eur. J. Soil Biol. 61(2014) 20-26. https://doi.org/10.1016/j.ejsobi.2013.12.008
- B.H. Ownley, D.G. Kimberly, E.V. Fernando, Endophytic fungal entomopathogens with activity against plant pathogens: ecology and evolution, BioControl 55 (2010) 113-128. https://doi.org/10.1007/s10526-009-9241-x

- B. Prapagdee, C. Kuekulvong, S. Mongkolsuk, Antifungal potential of extracellular metabolites produced by *Streptomyces hygroscopicus* against phytopathogenic fungi, Int. J. Biol. Sci. 4 (2008) 330. https://doi.org/10.7150/ijbs.4.330
- A. Raps, S. Vidal, Indirect effects of an unspecialized endophytic fungus on specialized plant-herbivorous insect interactions, Oecologia 114 (1998) 541–547.
- S. Ravnskov, B. Jensen, I.M. Knudsen, L. Bødker, D.F. Jensen, L. Karliński, et al., Soil inoculation with the biocontrol agent *Clonostachys rosea* and the mycorrhizal fungus *Glomus intraradices* results in mutual inhibition, plant growth promotion and alteration of soil microbial communities, Soil Biol. Biochem. 38 (2006) 3453-3462. https://doi.org/10.1016/j.soilbio.2006.06.003
- C. Sandberg, Evaluation of the antagonistic activity of *Rhodotorula babjevae* and its extracellular compounds towards postharvest pathogens, (2019)
- D.A. Schisler, M. Yoshioka, M.M. Vaughan, C.A. Dunla, A.P. Rooney, Nonviable biomass of biocontrol agent *Papiliotrema flavescens* OH 182.9 3C enhances growth of *Fusarium graminearum* and counteracts viable biomass reduction of *Fusarium* head blight, Biol. Control 128 (2019) 48-55. https://doi.org/10.1016/j.biocontrol.2018.09.006
- V. Shanthiyaa, D. Saravanakumar, L. Rajendran, G. Karthikeyan, K. Prabakar, T. Raguchander, Use of *Chaetomium globosum* for biocontrol of potato late blight disease, Crop Prot. 52 (2013) 33-38. https://doi.org/10.1016/j.cropro.2013.05.006
- K. Soytong, S. Kanokmedhakul, V. Kukongviriyapa, M. Isobe, Application of *Chaetomium* species (Ketomium) as a new broad spectrum biological fungicide for plant disease control, Fungal Divers 7 (2001) 1-15.
- J.H. Sun, J.J. Luo, L. Tian, C.L. Li, Y. Xing, Y. Shen, New Evidence for the Role of Ethylene in Strawberry Fruit Ripening, J.Plant Growth Regul. 32 (2013) 461–470. https://doi.org/10.1007/s00344-012-9312-6
- L. Szwajkowska-Michałek, H. Kwaśna, P. Łakomy, J. Perkowski, Inhibition of *Armillaria* and *Heterobasidion* growth by *Penicillium adametzii* isolated from *Pinus sylvestris* forest soil, Forest Pathol. 42 (2012) 454-466. https://doi.org/10.1111/j.1439-0329.2012.00780.x
- W. Yakti, G.M. Kovács, P. Vági, P. Franken, Impact of dark septate endophytes on tomato growth and nutrient uptake, Plant Ecol. Divers. 11 (2018) 637-648. https://doi.org/10.1080/17550874.2019.1610912

- J. Yin, F. Qian, J. Huang, New functions of *Ceriporia lacerate* in phytophthora blight control and growth promotion of eggplants, Sci. Agricul. Sin. 51 (2010) 2300-2310.
- S. Zhang, G. Yantai, X. Bingliang, Application of plant-growth-promoting fungi *Trichoderma longibrachiatum* T6 enhances tolerance of wheat to salt stress through improvement of antioxidative defense system and gene expression, Front. Plants Sci. (2016) 1405. https://doi.org/10.3389/fpls.2016.01405
- G. Zimmermann, Review on safety of the entomopathogenic fungi *Beauveria bassiana* and *Beauveria brongniartii*, Biocontrol Sci. Tec. 17 (2007) 553-596. https://doi.org/10.1080/09583150701309006